

Day 14 - Intermediate DSA : Bit Manipulations - 1

Fri, 10 Jun 2022

1 Goal 200 / 200

100% Class Attendance

5 / 5 Problems Solved

Session 100%

Assignment 2/2

Homework 3/3

Help Requests

Feedback

Question

Hints

Help Requests

Submissions

Discussions

Your Score: 200

Max Score: 200

Q2. Interesting Array

Solved



Stuck somewhere?

Ask for help from a TA and get it resolved.

Get help from TA.

Problem Description

You have an array **A** with **N** elements. We have **two** types of operation available on this array :

1. We can split an element **B** into two elements, **C** and **D**, such that $B = C + D$.

2. We can merge two elements, **P** and **Q**, to one element, **R**, such that $R = P \oplus Q$ i.e., XOR of P and Q.

You have to determine whether it is possible to convert array A to size 1, **containing a single element equal to 0** after several splits and/or merge?

Problem Constraints

1 <= N <= 100000

1 <= A[i] <= 10⁶

Input Format

The first argument is an integer array A of size N.

Output Format

Return "Yes" if it is possible otherwise return "No".

Example Input

Input 1:

A = [9, 17]

Input 2:

A = [1]

Example Output

Output 1:

Yes

Output 2:

No

Example Explanation

Explanation 1:

Following is one possible sequence of operations -
1) Merge i.e 9 XOR 17 = 24
2) Split 24 into two parts each of size 12
3) Merge i.e 12 XOR 12 = 0
As there is only 1 element i.e 0. So it is possible.

Explanation 2:

There is no possible way to make it 0.

See Expected Output

Java 8 (Array Support)

```
1 import java.util.Scanner;
2
3 public class Solution
4 {
5     public static void main(String[] args)
6     {
7         Scanner sc = new Scanner(System.in);
8         String[] input = sc.nextLine().split(" ");
9         int N = input.length;
10        int[] arr = new int[N];
11
12        for(int i = 0; i < N; i++)
13        {
```

Test Output Code's All Neat!

Compiling your Code...
> Success!

Running Test Cases...
> Test Case - Easy Success
> Test Case - Hard Success

Final Verdict

Test Test With Custom Input

Submit